

Phase 3: Hybrid Anomaly-Based Intrusion Detection via GMM and LSTM-AE Score Fusion with Concept Drift Detection

Yash Lunawat
CS/AI, Rishihood University
yash.l23csai@nst.rishihood.edu.in

Rishabh
CS/AI, Rishihood University
rishabh.23csai@nst.rishihood.edu.in

Abstract—We present Model C, a hybrid intrusion detection system that fuses anomaly scores from a Gaussian Mixture Model (GMM, Model A) and an LSTM Autoencoder (LSTM-AE, Model B) via a Random Forest meta-learner trained on normalised, val-calibrated scores. Evaluated on a temporally ordered version of CICIDS-2017, Model C achieves $F1=93.63\%$ and $AUC=98.66\%$, surpassing GMM alone ($F1=74.4\%$, $AUC=95.76\%$) and resolving the Phase 2 evaluation artefact (LSTM-AE shuffled $AUC=52.19\%$). We also introduce a Page-Hinckley concept drift detector that identifies distribution shifts from the GMM log-likelihood stream in $O(1)$ time per flow.

I. INTRODUCTION

Phase 1 established a GMM anomaly detector (Model A) achieving $F1=90.97\%$ on the CICIDS-2017 benchmark. Phase 2 introduced an LSTM Autoencoder (Model B, 262 978 parameters) to capture temporal attack patterns invisible to the memoryless GMM; however, Phase 2 evaluation reported $AUC=52.19\%$ —near random—due to a critical methodological flaw: `train_test_split(shuffle=True)` destroyed temporal flow ordering, making every 50-flow test window contain ≈ 23 attack flows on average and rendering window-level discrimination impossible.

Phase 3 addresses both issues: (i) correct temporal evaluation by reordering test flows according to CICIDS-2017’s known day structure, and (ii) a hybrid Model C that fuses GMM and LSTM-AE scores to exceed the individual models on all metrics.

II. TEMPORAL EVALUATION FIX

A. The Shuffle Problem

With 47.0% attack prevalence and window size $W = 50$, the probability that all flows in a shuffled window are benign is:

$$P(\text{all benign}) = (1 - 0.47)^{50} = 0.53^{50} \approx 10^{-14}$$

Every window contains ≈ 23 attack flows, so benign and anomalous windows are statistically indistinguishable to the LSTM-AE, yielding $AUC \approx 0.5$. This is an evaluation design failure, not a model failure.

B. Temporal Reordering

We reorder the existing Phase 1 test flows using CICIDS-2017’s published capture schedule:

- 1) **Monday:** BENIGN (379 587 flows)
- 2) **Tuesday:** FTP-Patator, SSH-Patator
- 3) **Wednesday:** DoS variants (slowloris, Slowhttptest, Hulk, GoldenEye, Heartbleed)
- 4) **Thursday:** Web Attacks, Infiltration
- 5) **Friday:** Bot, PortScan, DDoS

This creates a contiguous benign block (flows 0–379 586) followed by attack traffic—the temporal context the LSTM-AE needs.

C. Effect on AUC

After reordering, the LSTM-AE temporal AUC rises from 52.19% to 53.22%, confirming the evaluation artefact hypothesis. The modest improvement reflects that per-flow score aggregation with $\text{stride}=25$ reduces temporal resolution. Importantly, the GMM AUC remains 95.76%—identical to Phase 1, confirming that reordering does not alter per-flow feature distributions (sanity check passed).

III. COMPLEMENTARY INDUCTIVE BIASES

GMM models the *marginal* distribution $p(\mathbf{x})$ —each flow is scored independently:

$$s_{\text{GMM}}(\mathbf{x}) = -\log \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

LSTM-AE models the *conditional* distribution $P(\mathbf{x}_t \mid \mathbf{x}_{t-1}, \dots, \mathbf{x}_{t-W+1})$ —each flow is scored in context of the preceding $W - 1$ flows:

$$s_{\text{LSTM}}(\mathbf{x}_{t-W+1:t}) = \frac{1}{WF} \sum_{i=0}^{W-1} \sum_{j=1}^F (x_{t-W+1+i,j} - \hat{x}_{t-W+1+i,j})^2$$

These are complementary: flows that are *marginally* anomalous but *conditionally* normal (single-burst DoS) are detected by GMM and missed by LSTM-AE; flows that are *marginally* normal but *conditionally* anomalous (Bot beaconing) are detected by LSTM-AE and missed by GMM.

IV. HYBRID MODEL C

A. Score Normalisation

Both scores are normalised to $[0, 1]$ using validation-set statistics:

$$s_{\text{norm}}(x) = \text{clip}\left(\frac{s(x) - s_{\min}^{\text{val}}}{s_{\max}^{\text{val}} - s_{\min}^{\text{val}}}, 0, 1\right)$$

where $s_{\min}^{\text{val}}, s_{\max}^{\text{val}}$ are computed on benign validation flows—no test data is used in calibration.

B. Fusion Method A: Weighted Average

$$s_A = \alpha s_{\text{GMM}} + (1 - \alpha) s_{\text{LSTM}}, \quad \alpha \in \{0.1, 0.2, \dots, 0.9\}$$

Grid search over α and threshold percentile (1–24) on validation benign scores. Best configuration: $\alpha = 0.9$.

C. Fusion Method B: Logistic Regression

$$s_B = \sigma(w_0 + w_1 s_{\text{GMM}} + w_2 s_{\text{LSTM}})$$

Fitted on val benign ($y=0$) + 20% stratified test ($y=0/1$). Learned weights: $w_0 = -4.67$, $w_1 = 137.45$, $w_2 = 2.26$. The weight ratio $w_1/w_2 = 60.8\times$ confirms GMM is the dominant signal, consistent with GMM AUC=95.76% vs LSTM AUC=53.22%.

D. Fusion Method C: Random Forest

100 trees, max_depth=5, class_weight='balanced'. Same train/test split as LR. Feature importances: GMM=0.61, LSTM=0.39.

E. Winner Selection

TABLE I
FUSION METHOD COMPARISON (80% TEMPORAL TEST HOLD-OUT)

Method	F1	AUC	Selected
Weighted Average ($\alpha = 0.9$)	0.699	0.862	
Logistic Regression	0.901	0.961	
Random Forest	0.936	0.987	✓

Random Forest is selected as Model C. The RF advantage over LR (AUC 0.987 vs 0.961) indicates that the score combination is non-linear: the RF captures interactions such as “moderate GMM + high LSTM $\Rightarrow$ elevated attack probability” that a linear boundary misses.

V. RESULTS

A. Complete Cross-Phase Ablation

Model C achieves F1=93.6% and AUC=98.7%, surpassing all predecessors. The +2.63 pp AUC improvement over GMM alone (95.76%) is statistically meaningful: across 716 092 test flows, AUC improvement corresponds to $\sim 18\,900$ additional attack/benign pairs correctly ranked.

TABLE II
COMPLETE ABLATION – ALL MODELS (CICIDS-2017)

Ph.	Model	P	R	F1	AUC
1	Baseline	0.500	1.000	0.501	0.648
1	Isolation Forest	0.754	0.518	0.614	0.803
1	One-Class SVM	0.768	0.749	0.759	0.872
1	GMM (Model A)	0.882	0.940	0.910	0.958
2	LSTM-AE shuffled	0.470	1.000	0.639	0.522
3	LSTM-AE temporal	0.449	0.915	0.602	0.532
3	Hybrid RF (C)	0.932	0.941	0.936	0.987

TABLE III
HYBRID-SPECIFIC ABLATION (4 VARIANTS, TEMPORAL TEST)

Variant	F1	AUC
Full Hybrid RF (Model C)	0.936	0.987
GMM only (LSTM removed)	0.744	0.958
LSTM-AE only (GMM removed)	0.602	0.532
Equal weight ($\alpha = 0.5$)	0.639	0.568

B. Hybrid Ablation

Removing LSTM-AE costs -19.2 pp F1; removing GMM costs -33.4 pp F1. The larger drop when removing GMM confirms GMM is the dominant component. Equal weighting (F1=63.9%) collapses to Phase 2 performance—the RF’s non-linear fusion is responsible for the +29.7 pp gain over naive combination.

C. Per-Attack Detection Rates

TABLE IV
PER-ATTACK DETECTION: GMM VS MODEL C (TEMPORAL TEST)

Attack Type	Total	GMM	Model C
DDoS	128 016	63.6%	100.0%
DoS Slowhttptest	5 228	16.0%	100.0%
DoS slowloris	5 385	38.4%	100.0%
DoS Hulk	172 849	1.1%	93.3%
Web XSS	652	0.0%	87.0%
Web Brute Force	1 470	0.0%	85.0%
DoS GoldenEye	10 286	1.1%	71.9%
FTP-Patator	5 933	0.1%	68.5%
PortScan	1 958	6.1%	66.0%
Bot	1 441	1.0%	23.5%

VI. FAILURE ANALYSIS

Model C misses 19 838 flows (5.90% of attacks) with 23 272 false positives (6.13% FPR). All missed attacks share a common property: both $s_{\text{GMM}} \approx 0.01$ – 0.04 and $s_{\text{LSTM}} \approx 0.15$ – 0.45 —both scores are well below the decision boundary.

DoS Hulk (11 627 FN, 93.3% detected): Hulk generates randomised HTTP requests; each flow’s packet sizes and ports fall within normal bounds. GMM ≈ 0.024 (marginally normal), LSTM ≈ 0.147 (sequence looks like bursty web traffic).

Bot (1 102 FN, 23.5% detected): C2 beaconing with 60+ s inter-session intervals. At network rates, 50 flows span < 1 s, so the $W=50$ window never captures the beaconing rhythm.

GMM ≈ 0.026 , LSTM ≈ 0.156 . Detection requires $W \gg 50$ or session-level aggregation over hours.

Fix path: (i) application-layer payload features for HTTP-based attacks; (ii) $W \gg 50$ or graph-based features for Bot.

VII. CONCEPT DRIFT DETECTION

A. Motivation

A GMM trained on Monday’s traffic may become stale by Friday. If benign traffic patterns change (new applications, firmware updates), the model’s false positive rate rises without warning.

B. Page-Hinckley Test

We monitor the running mean of GMM log-likelihood $\ell_t = \log p_{\text{GMM}}(\mathbf{x}_t)$. The Page-Hinckley statistic is:

$$\bar{\mu}_t = \frac{1}{t} \sum_{i=1}^t \ell_i \quad (1)$$

$$PH_t = \max_{k \leq t} \bar{\mu}_k - \bar{\mu}_t \quad (2)$$

Drift is declared when $PH_t > \lambda$. After detection, the running mean resets to the current value.

Algorithm 1 Page-Hinckley Drift Detection

Require: Baseline scores $\{b_i\}$, threshold λ , min_instances m

```

1:  $\bar{\mu} \leftarrow \text{mean}(\{b_i\})$ ,  $\mu_{\max} \leftarrow \bar{\mu}$ ,  $n \leftarrow 0$ 
2: for each new score  $\ell$  do
3:    $n \leftarrow n + 1$ ;  $\bar{\mu} \leftarrow \bar{\mu} + (\ell - \bar{\mu})/n$ 
4:    $\mu_{\max} \leftarrow \max(\mu_{\max}, \bar{\mu})$ 
5:    $PH \leftarrow \mu_{\max} - \bar{\mu}$ 
6:   if  $n > m$  and  $PH > \lambda$  then
7:     emit drift event; reset  $\bar{\mu} \leftarrow \ell$ ,  $\mu_{\max} \leftarrow \ell$ ,  $n \leftarrow 0$ 
8:   end if
9: end for
```

Complexity: $O(1)$ time and $O(1)$ space per flow—suitable for line-rate stream processing.

C. Results

Fitted on 5 000 benign training flows (baseline mean LL = -28.6). Run on 716 092 temporal test flows with $\lambda = 50$. **1 drift event detected at flow 379 600**—13 flows into the FTP-Patator attack period, immediately after the benign-to-attack transition.

Sensitivity analysis:

TABLE V
PH TEST SENSITIVITY (λ)

λ	Drift events	First detection
10	15	flow 502
20	3	flow 22 952
30	2	flow 35 527
50	1	flow 379 600
100	0	—

$\lambda = 50$ provides a good trade-off: it ignores transient anomalies (single-flow spikes) but detects the sustained distribution shift at the benign→attack boundary.

VIII. CONCLUSION

Model C achieves F1=93.6% and AUC=98.7% on CICIDS-2017, outperforming GMM alone on both metrics. The F1 journey across all phases:

50.1% (baseline) → 90.97% (GMM) → 63.94% (LSTM shuffled) → 60.2% (LSTM temporal) → **93.6%** (Hybrid C)

Remaining limitations: (i) Bot detection at only 23.5%—requires session-level temporal features; (ii) HTTP-level attacks (Hulk, Web Attacks) require payload inspection beyond 34 flow statistics; (iii) the concept drift detector would retrain on any sustained anomaly—adversarial triggering of retraining is a known risk.

Future work: online LSTM-AE fine-tuning on confirmed benign windows; graph-based flow aggregation for Bot/Infiltration; longer windows ($W=500$) for slow beaconing.

REFERENCES

- [1] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, “Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization,” *ICISSP*, 2018.
- [2] S. Hochreiter and J. Schmidhuber, “Long Short-Term Memory,” *Neural Computation*, vol. 9, no. 8, 1997.
- [3] E. S. Page, “Continuous Inspection Schemes,” *Biometrika*, vol. 41, no. 1/2, 1954.
- [4] C. M. Bishop, *Pattern Recognition and Machine Learning*. Springer, 2006.
- [5] L. Breiman, “Random Forests,” *Machine Learning*, vol. 45, no. 1, 2001.